

CSC15S Assignment 3 – Cybersec Laws and Linux File Systems

Total: 100pts

What to turn in: zip and upload all your related files to **Assignment 3** in Blackboard. Your zip should include the following files:

- (1) CSC15Slab3.docx (with all questions completed)
- (2) log_Mission1.txt

Part 1. Security Models

[20 pts. Estimated time to complete: 30 minutes]

Finish the reading and answer the following questions (up to 100 words each).

1. What are the major limitations of Bell-LaPadula? [10 pts]

The Bell-LaPadula model is focused on data confidentiality and doesn't address data integrity or availability. It enforces no read up, no write down meaning users cannot access higher classification levels or write to lower levels. However, it does not prevent unauthorized modification and lacks consideration for covert channels, where information can be leaked indirectly. Additionally, it is overly rigid for real-world applications requiring more flexibility, such as collaborative environments or systems that need dynamic access control.

2. In Linux, as the OS designer, how would you enforce the Bell-LaPadula? [10 pts]

To enforce Bell-LaPadula in Linux, Mandatory Access Control mechanisms like SELinux (Security-Enhanced Linux) can be used. SELinux provides policies that control access based on sensitivity levels. By defining domains for processes and sensitivity labels for files, you can restrict access according to the Bell-LaPadula "no read up, no write down" policy. High-security processes would only be able to read or write data at or below their level, ensuring confidentiality. Custom policies can be configured to enforce these access control rules across users and system processes.

Part 2. Dr. Evil's Training Room. Mission 1 – The Hell of Linux Files.

[80 pts. Estimated time to complete: 1.5 hrs]

Dr. Evil has prepared some nice “appetizers” for you. It’s an automated bot grader and trainer. Let’s download it first:

- (1) Start a browser in your CSC15S VM Image and visit Blackboard class account
- (2) Navigate to Assignments -> Assignment 3
- (3) Right click on mission_files and download. You can save it to the Downloads folder.

Then you can start the program following the commands below:

```
cd ~/Downloads
chmod 755 mission_files
sudo ./mission_files
```

You need to start another terminal to complete the training. Note that we are expecting you as a highly efficient system administrator. So all your actions are timed. If it takes too long to complete a challenge, there will be penalty. But fortunately, you are allowed to redo any challenge as many times as you like.

Once you feel ready to submit, please attach log_Mission1.txt to Blackboard. The score report should look like the one shown in Figure below. Please do not tamper the file because it is checksummed by EvilTrainer.

```
4cS220aa44a469137454e810e1b9d73f0d1
=====
Mission1: Hell of Linux Files (80 pts)
Name: Sean Fu
Grade: 28.50 pts
Time Spent: 208 seconds
=====
QID  Title                Pts    Time    QID  Title                Pts    Time
-----
1    Create Dir            0.0/5   6        2    Simple List           5.0/5   21
3    Dir Navigation        0.0/10  20       4    Kill Files            6.0/10  35
5    Make Copies           4.50/15 25       6    Move Files            0.0/10  46
7    Set Perimssions       3.0/15  32       8    Run Shell Script      10.0/10 23
-----
Enter question ID (e.g., 1) to redo a question or hit Q to quit
%13792012345-----
0%139209234----
6%139209234----
```